

Prescribed Banzhaf power: closed-form games, congruence conditions, and certified weighted obstructions

Yongjin Chen^a, Ziyang Xue^b, Songze Zhu^{a,*}

^a School of Public Security Management, People's Public Security

University of China, Beijing 100038, China

^b School of Investigation, People's Public Security University of China,

Beijing 100038, China

* Corresponding author. E-mail: 20196846@ppsuc.edu.cn

E-mail address: 15620821066@163.com (Z. Xue)

Abstract

Designing a voting rule that realizes a prescribed distribution of voting power, the inverse power index problem, is a fundamental computational task for parliaments, councils, and shareholder assemblies. Kurz's (2015) egalitarian benchmark, in which one player holds half the power of each of the others, had been settled only by computation, up to eighteen voters, with no construction valid for every number of voters. This paper develops an exact, reproducible computational methodology for the inverse Banzhaf problem and applies it to that benchmark.

On the constructive side, one parametric family of simple games (the W-family) with an exact closed-form swing formula yields explicit realizing games for every number of voters from six upward, resolving Kurz's Conjecture 16 and correcting the sole erroneous published example. The same analysis extends to a two-parameter target family in which one player's target share differs from the rest, whose realization by this family is decided by an explicit integrality condition, the numbers of voters it covers forming a union of residue classes

with a common period. A cubic-variable compact formulation solves every instance up to sixty-four voters in seconds to minutes and up to eighty within available memory. On the obstruction side, a sharpened swap/block rigidity condition drives a deterministic certification, by constraint programming, that no weighted majority game on six to ten voters realizes the target, at a conclusive weight bound. The certified optimal weighted approximation is exact for up to eight voters.

A case study on the historical weighted Council of the European Union reproduces its exact Banzhaf distribution, shows it far from the target, and improves the distance about threefold under the certified optimal redesign for six voters. Every numerical claim is reproducible from the companion code package.

Keywords: simple games; Penrose–Banzhaf power index; weighted majority games; Diophantine conditions; constraint programming

1. Introduction: the inverse Banzhaf problem

Designing a voting rule for a committee of *dissimilar* members, such as a weighted parliament, the Council of the European Union, or a federal upper chamber, is a decision problem: the designer must decide how voting *power* is to be distributed among members. The Banzhaf index (and the related Shapley–Shubik index) quantify power as the normalized number of *swings*, i.e., coalitions in which a member is decisive (Penrose 1946; Felsenthal & Machover 1998); for a weighted majority game, computing these indices is NP-hard (Matsui & Matsui 2001). The **inverse power index problem** is the formal dual of this computation: given a desired power distribution, decide whether a voting rule realizes it, construct one if so, and certify impossibility otherwise (Nurmi 1982; Kurz & Napel 2014; de Keijzer et al. 2014). Abstractly the question is one of realizability. The index is a map from a finite combinatorial class (all simple games on n voters, or the subclass of weighted majority games that real voting bodies use; Muroga 1971) onto the rational points of a simplex, and one asks which points lie in its image, by which constructions, and at what cost. For the graph games studied here the constructive half reduces to a linear Diophantine condition together with the graphicality of a degree sequence, so the inverse problem takes an arithmetic shape: which n , and which target ratios, satisfy it.

This paper treats both halves exactly. The primary target has resisted a unified treatment since 2014: the vector $\psi^n = (2, \dots, 2, 1) / (2n - 1)$, (1) which assigns

to $n - 1$ members twice the power of the remaining member. It is the case $(a, b) = (2, 1)$ of the two-parameter family $\psi^n(a, b) \propto (a, \dots, a, b)$, for which Section 3 gives a closed-form existence criterion (Theorem 2). Kurz (2015) conjectured the existence of a realizing simple game for every $n \geq 6$:

Conjecture 16 (Kurz 2015). For every $n \geq 6$ there exists a simple game v on n voters such that $Bz(v) = \psi^n$, where Bz is the (normalized) Penrose–Banzhaf index.

and the natural dual, that no *weighted majority* game can do so (Kurz’s Conjecture 15). Kurz and Napel (2014) verified Conjecture 16 computationally for $6 \leq n \leq 18$, but gave no construction valid for all n ; we provide one (Corollaries 14–17, Section 11), and settle the weighted obstruction computationally for $n \leq 10$ (Section 9).

The paper is organized as an exact computational study with two complementary halves. On the constructive side, Sections 3–7 give closed-form families, exact swing formulas, and explicit games realizing ψ^n for every $n \geq 6$, together with the generalization of the complement-of-edges criterion to the whole family $\psi^n(a, b)$ (Theorem 2, Section 3). Section 8 summarizes the constructive half. On the obstruction side, Section 9 derives a sharp structural rigidity property (M') and certifies by the constraint-programming satisfiability (CP-SAT) solver that no weighted majority game realizes ψ^n for $n \leq 10$. Section 10 collects the two engines as a reusable methodology, with complexity statements, instance tables, and a benchmark of three formulations on a common instance set. Section 11 develops the unified W-family framework, the exact closed-form constructions, and the open problems. Section 12 applies the methodology as an exact computational case study to the historical weighted Council of the European Union, whose actual Banzhaf power distributions fall exactly in the certified range. Section 13 concludes.

2. Preliminaries

Let $N = \{1, \dots, n\}$ be a set of voters. A **simple game** (Taylor & Zwicker 1999) is a monotone Boolean function $v : 2^N \rightarrow \{0, 1\}$ with $v(\emptyset) = 0$ and $v(N) = 1$. A coalition $S \subseteq N$ is **winning** if $v(S) = 1$ and **losing** otherwise. A winning coalition is **minimal winning** if all its proper subsets are losing; a simple game is uniquely determined by its family of minimal winning coalitions.

For $i \in N$, let $\beta_i(v) = |\{S \subseteq N \setminus \{i\} : v(S) = 0, v(S \cup \{i\}) = 1\}|$ be the number of **swings** of player i (the Banzhaf measure). The (normalized) **Penrose–Banzhaf index** is $Bz_i(v) = \beta_i(v) / \sum_j \beta_j(v)$. Thus $Bz(v) = \psi^n$ iff $\beta(v)$ is proportional to $(2, \dots, 2, 1)$, i.e., iff $n - 1$ players have an equal number $2c$ of swings and one player has c swings, for some integer $c \geq 1$.

3. The complement-of-edges family

For a graph G on vertex set N with edge set E , consider the simple game v_G whose minimal winning coalitions are exactly the complements of the edges of G : $v_G(S) = 1 \iff S \supseteq N \setminus \{a, b\}$ for some edge $\{a, b\} \in E$. Equivalently, a coalition is winning iff the voters outside it are contained in some edge of G . We prove:

Theorem 1. Let G be a graph on n vertices with edge set E . Suppose (i) G has no isolated vertices, and (ii) G is not a star (no vertex is incident with every edge). Then the Banzhaf swing counts of v_G are $\beta_i(v_G) = |E| + (n - 1) - 2 \cdot \deg_G(i)$ ($i \in N$). (2)

Proof. Player i has a swing on a losing coalition S with $i \notin S$ and $S \cup \{i\}$ winning. The latter means $S \cup \{i\} \supseteq N \setminus \{a, b\}$ for some edge $\{a, b\}$. - If $|S| = n - 3$, then $S \cup \{i\} = N \setminus \{a, b\}$ is a $(n - 2)$ -set, so $S = N \setminus (\{a, b\} \cup \{i\})$ with $i \notin \{a, b\}$; such S is losing since it has size $n - 3$. Each edge not containing i contributes exactly one such swing, giving $|E| - \deg_G(i)$ swings. - If $|S| = n - 2$, then $S \cup \{i\} = N \setminus \{x\}$ for some $x \neq i$. This is winning iff $\deg_G(x) \geq 1$, which holds by (i). The set $S = N \setminus \{x, i\}$ is losing iff $\{x, i\} \notin E$ (otherwise S contains the winning $(n - 2)$ -set $N \setminus \{x, i\}$). Hence exactly $(n - 1) - \deg_G(i)$ swings of this type. - If $|S| = n - 1$, then $S = N \setminus \{i\}$ and $S \cup \{i\} = N$ is winning. But $N \setminus \{i\}$ is losing iff every edge contains i , i.e., G is a star centered at i , excluded by (ii). Hence no swings of this type. - No other cardinality yields a swing.

Summing gives $\beta_i(v_G) = |E| + (n - 1) - 2 \cdot \deg_G(i)$. ■

Corollary 2. If G has all vertices but one of common degree d and the exceptional vertex of degree $d + c/2$, and if $|E| + (n - 1) - 2d = 2c$, then $Bz(v_G) = \psi^n$. Since $2|E| = nd + c/2$, the existence of such a graph is equivalent to the solvability of $7c = d(2n - 8) + 4(n - 1)$ (3) in positive integers d, c with $1 \leq d \leq n - 2$ and $d + c/2 \leq n - 1$, together with the graphicality of the degree

sequence $(d, \dots, d, d+c/2)$ (Erdős & Gallai 1960).

For $6 \leq n \leq 18$, equation (3) admits such a solution exactly for $n \in \{6, 7, 9, 13, 14, 16, 17\}$. For $n \equiv 4 \pmod{7}$ it has no solution at all; at $n = 8$ its only solution has $d = 0$, which the absence of isolated vertices forbids (the degree sequence is not graphical); and at each of $n = 10, 12$ and 15 the exceptional degree of the unique admissible solution exceeds $n - 1$. Hence the complement-of-edges family alone does **not** cover all n , which is consistent with the conjecture requiring a genuinely different construction for the remaining values.

The condition (3) is solvable for *infinitely many* n : for $n \leq 30$ the solvable values are 6, 7, 9, 13, 14, 16, 17, 20, 21, 23, 24, 27, 28, 30, ... Indeed, the residue classes $n \equiv 6, 0, 2, 3 \pmod{7}$ are solvable for every admissible n with the *constant* common degree $d = 2, 3, 1, 4$ respectively, since then $\deg_G(n) = (4n+4)/7, (5n+7)/7, (3n+1)/7, (6n+10)/7 \leq n - 1$. Thus Conjecture 16 holds, by Theorem 1, for **four of the seven residue classes** (i.e. for infinitely many n); the values $n \equiv 1, 4, 5 \pmod{7}$ require a genuinely different construction (given in Sections 4 and 9).

Beyond the single target: the family $\psi^n(a, b)$

The construction above answers (1) for one target. It extends to a two-parameter family of targets at no extra cost in machinery. For integers $a > b \geq 1$ put

$$\psi^n(a, b) = (a, \dots, a, b)/((n-1)a + b),$$

so that one distinguished player holds $b/((n-1)a + b)$ of the total power and each of the other $n - 1$ players holds $a/((n-1)a + b)$. The target of Sections 3 to 9 is the case $(a, b) = (2, 1)$, for which $(n-1)a + b = 2n - 1$.

Theorem 2. Let $n > 4$ and let $a > b \geq 1$ be integers. Suppose some integer $t \geq 1$ satisfies

- (i) $c = t(a - b)$ is even, and $d = [t(3a + b)/2 - 2(n - 1)]/(n - 4)$ is an integer with $1 \leq d \leq n - 2$;
- (ii) $d + c/2 \leq n - 1$; and
- (iii) the sequence $(d, \dots, d, d + c/2)$ is the degree sequence of a graph G with no isolated vertex and not a star.

Then $Bz(v_G) = \psi^n(a, b)$.

Proof. Let $D = d + c/2$ be the degree of the distinguished vertex. By Theorem 1, $\beta_i(v_G) = |E| + (n - 1) - 2 \deg_G(i)$, so for every $i \neq n$

$$\beta_i - \beta_n = 2(D - d) = c.$$

The gap between the power of an ordinary player and the power of the distinguished player is therefore c , whatever else is true of G . Demanding $\beta_i = ta$ and $\beta_n = tb$ for a common scale t forces $c = t(a - b)$, which is (i). The handshake lemma applied to the degree sequence gives $2|E| = (n - 1)d + D = nd + c/2$; substituting this and $\beta_i = ta$ into Theorem 1 gives

$$nd/2 + c/4 + (n - 1) - 2d = ta,$$

which on multiplying by 4 and using $c = t(a - b)$ becomes $2d(n - 4) = t(3a + b) - 4(n - 1)$; that is the expression for d in (i). Conditions (ii) and (iii) are precisely what is needed for the sequence to be realizable and for Theorem 1 to apply to the realizing graph. Then $\beta_i = ta$ for $i \neq n$ and $\beta_n = tb$, so $\beta(v_G) = t((n - 1)a + b) \cdot \psi^n(a, b)$, and normalizing gives $Bz(v_G) = \psi^n(a, b)$. ■

At $(a, b) = (2, 1)$ Theorem 2 is not a new result. There $c = t$, and $2d(n - 4) = 7t - 4(n - 1)$ rearranges to equation (3). Corollary 2 is the case $(a, b) = (2, 1)$ of Theorem 2, and the derivation above reproduces it independently.

Which targets this family covers. Because $c \geq 2$ in every admissible instance (an odd c would make $d + c/2$ non-integral) we have $D > d$, so the bound $d \leq n - 2$ is implied by (ii) rather than assumed alongside it. Substituting it into the expression for d bounds the range of t by

$$t \leq 2[(n - 4)(n - 2) + 2(n - 1)]/(3a + b),$$

a quantity below 4000 for every $a > b \geq 1$ and every $n \leq 120$. The search below is therefore exhaustive over t , and the table is a statement about the reach of the family rather than a report of where a search happened to stop.

(a,b)	3a+b	smallest $n \leq 120$	
		realized by this family	covered residues mod 3a+b
(2,1)	7	6, 7, 9, 13, 14, 16, 17, 20, 21, 23, 24, 27, ...	4 of 7: 0, 2, 3, 6

(a,b)	3a+b	smallest $n \leq 120$	
		realized by this family	covered residues mod 3a+b
(3,1)	10	7, 10, 12, 15, 17, 20, 22, 25, 27, 30, 32, 35, ...	4 of 10: 0, 2, 5, 7
(3,2)	11	8, 11, 13, 14, 16, 17, 19, 22, 24, 25, 27, 28, ...	8 of 11: 0, 2, 3, 5, 6, 7, 8, 10
(4,1)	13	9, 15, 22, 28, 35, 41, 48, 54, 61, 67, 74, 80, ...	2 of 13: 2, 9
(4,2)	14	6, 7, 9, 13, 14, 16, 17, 20, 21, 23, 24, 27, ...	8 of 14: 0, 2, 3, 6, 7, 9, 10, 13
(4,3)	15	7, 8, 10, 12, 13, 15, 16, 17, 18, 20, 22, 23, ...	12 of 15

Three features of the table are worth recording. First, coverage is a property of the *ratio* $a:b$ and not of the pair: $(4,2)$ and $(2,1)$ denote the same target, and the covered set for $(4,2)$ is exactly the preimage modulo 14 of the covered set for $(2,1)$ modulo 7. Second, the density of coverage varies sharply with the ratio, from 2 of 13 classes at $(4,1)$ to 12 of 15 at $(4,3)$. The question “for which ratios $a:b$ does this family cover almost every n ?” is the natural quantitative form of the inverse problem beyond the single target, and we do not settle it here. Third, Theorem 2 does **not** dissolve the residue obstruction. At $(2,1)$ the family covers four of the seven classes, exactly as Corollary 2 and equation (3) said, and the remaining three still require the separate constructions of Sections 4 and 9. The generalization relocates the target; it does not remove the difficulty. A second route to the same family, the triangle construction, is denser: at the ratio 3:1 it reaches every $n \leq 80$ where the family above reaches four of ten residue classes, and the two are compared in Section 9.

Verification. Theorem 2 is proved above, so the computations that follow corroborate it rather than establish it. Every entry in the table with $n \leq$

18 was realized by an explicit graph built through Havel–Hakimi (Havel 1955; Hakimi 1962) and confirmed by enumerating all 2^n coalitions and counting swings. There were 36 realizations and no failures, and the $(2, 1)$ row reproduces the seven values that Section 3 reports independently from equation (3). For $18 < n \leq 120$ the degrees are still realized explicitly and the swing vector is recomputed by a second, polynomial-time derivation of Theorem 1 that does not use its case split on $|S|$; the game definition itself is checked by exhaustive enumeration only for $n \leq 18$. Both the constructions and the bound on t are reproduced in the companion code package.

4. The threshold-with-exceptions family

The complement-of-edges family does not cover every n . For the remaining small values a second, equally explicit family is useful. Let E be a set of pairs and F a set of triples of $[n]$. Define the game $v(E, F)$ by $v(S) = 1 \iff |S| \geq n - 1$, or $(|S| = n - 2 \text{ and } N \setminus S \notin E)$, or $(|S| = n - 3 \text{ and } N \setminus S \in F)$. (4) That is, the game is “almost” a majority-style threshold at size $n - 2$, with a set of *holes* ($N \setminus S \in E$) removed at level $n - 2$ and a set of *exceptions* ($N \setminus S \in F$) added at level $n - 3$.

Lemma (monotonicity). $v(E, F)$ is monotone (hence a simple game) iff no edge of E is contained in a triple of F , i.e. $E \cap \text{Pairs}(F) = \emptyset$, where $\text{Pairs}(F)$ is the set of all two-element subsets of triples in F . Indeed, a winning $(n - 3)$ -set $N \setminus X$ ($X \in F$) has a losing $(n - 2)$ -superset $N \setminus \{a, b\}$ exactly when $\{a, b\} \in E$ with $\{a, b\} \subseteq X$.

Theorem 3. For any E, F with $E \cap \text{Pairs}(F) = \emptyset$, the swing counts of $v(E, F)$ are $\beta_i(v(E, F)) = C(n - 1, 2) + |F| - |E| + 2 \cdot \deg_E(i) - 2 \cdot r_i + e_i$, (5) where $\deg_E(i)$ is the degree of i in the graph E , $r_i = |\{X \in F : i \in X\}|$, and $e_i = |\{X \in F : i \in X, X \setminus \{i\} \in E\}|$.

Proof. Player i swings on a losing S with $S \cup \{i\}$ winning. Enumerate by $|S|$:

- $|S| = n - 2$. Then S is losing, so $N \setminus S \in E$; the unique losing $(n - 2)$ -set with $i \notin S$ and $N \setminus S \in E$ requires $N \setminus S \ni i$, giving $\deg_E(i)$ swings.
- $|S| = n - 3$. Here $S = N \setminus \{i, a, b\}$ with $\{i, a, b\} \notin F$ and $\{a, b\} \notin E$, giving $C(n - 1, 2) - |E| + \deg_E(i) - r_i + e_i$ swings.
- $|S| = n - 4$. Here $S = N \setminus (X \cup \{i\})$ for $X \in F$ with $i \notin X$, giving $|F| - r_i$ swings.

- $|S| \leq n - 5$. Then $S \cup \{i\}$ has size at most $n - 4$ and is losing, so it yields no swing.

These cases exhaust the possible sizes of S , and summing the first three yields (5). ■

As an independent check, the identity was verified exhaustively against direct computation on random instances for $n = 8, 10, 12$.

Corollary 3. If $\deg_E(i) - r_i$ is constant, say D , for all $i \neq n$ and $\deg_E(n) - r_n = D - c/2$, then $\text{Bz}(v(E, F)) = \psi^n$ with scale $c = (n \cdot C(n - 1, 2) + (n - 6)|F| - (n - 4)|E|)/(2n - 1)$ (when this is a positive integer). In particular, such games realize ψ^n for every $n \in \{8, 10, 11, 12, 15\}$ not covered by Corollary 2 (found by solving (5) via mixed-integer programming; see Section 7), while for $n = 18$ a solution was obtained by Kurz and Napel.

5. Explicit games realizing ψ^n

For each $n \in \{6, 7, 9, 13, 14, 16, 17\}$ we give the graph G_n (edges written as pairs) such that $\text{Bz}(v_{G_n}) = \psi^n$; the minimal winning coalitions of v_{G_n} are the complements of these edges. The resulting swing vector is $\beta = c \cdot (2, \dots, 2, 1)$, displayed as a multiple of c .

n	graph G_n	c	$\beta(v_{G_n})$
6	$C_4(1 - 2 - 6 - 3 - 1) \cup K_3(4, 5, 6)$	4	(8, 8, 8, 8, 4)
7	C_6 on $\{1, \dots, 6\} \cup \text{star centered at } 7$	6	(12, $\dots$, 12, 6)
9	star at 9 on $\{1, 2, 3, 4\} \cup \text{matching } \{5, 6\}, \{7, 8\}$	6	(12, $\dots$, 12, 6)
13	star at 13 on $\{1, \dots, 8\} \cup \text{matching on } \{1, \dots, 8\} \cup C_4$ on $\{9, \dots, 12\}$	12	(24, $\dots$, 24, 12)
14	C_{13} on $\{1, \dots, 13\} + \text{chord } \{6, 10\} \cup \text{star at } 14 \text{ except } \{6, 10\}$	16	(32, $\dots$, 32, 16)
16	star at 16 on $\{1, \dots, 7\} \cup \text{matching on } \{8, \dots, 15\}$	12	(24, $\dots$, 24, 12)
17	star at 17 (to all) $\cup C_{16}$ on $\{1, \dots, 16\} \cup \text{matching } \{i, i + 8\} (1 \leq i \leq 8)$	24	(48, $\dots$, 48, 24)

Each entry is verified directly by computing the swing counts. For instance,

the $n = 6$ entry is the game whose minimal winning coalitions are $\{1, 2, 3, 4\}$, $\{1, 2, 3, 5\}$, $\{1, 2, 3, 6\}$, $\{1, 2, 4, 5\}$, $\{1, 3, 4, 5\}$, $\{2, 4, 5, 6\}$, $\{3, 4, 5, 6\}$ (the complements of the seven edges of $C_4 \cup K_3$), whose swing counts are $(8, 8, 8, 8, 8, 4)$.

6. A correction to the published example

Kurz and Napel (2014, footnote 19) state that the game on six voters whose minimal winning coalitions are $\{2, 4, 5, 6\}$, $\{2, 3, 4, 5\}$, $\{1, 3, 5, 6\}$, $\{1, 3, 4, 5\}$, $\{1, 2, 4, 6\}$, $\{1, 2, 3, 5\}$ attains the PBI vector $(8, 8, 8, 8, 8, 4)/44 = \psi^6$. This is **incorrect**: a direct computation (or a routine check by hand for, e.g., player 6, who has five swings) shows the swing counts are $\beta = (7, 7, 7, 7, 9, 5)$, $\sum \beta = 42$, so that $\text{Bz}(v) = (7, 7, 7, 7, 9, 5)/42 \neq \psi^6$. The claim that an exact solution exists for $n = 6$ is nevertheless true: the game in Section 5 (seven four-sets) attains ψ^6 . (An exhaustive enumeration over all simple games on six voters yields 360 games, in two symmetry classes, whose Banzhaf index equals ψ^6 . The enumeration covers 7,828,354 antichains of subsets, equivalently all monotone Boolean functions, of which 7,828,352 are simple games.)

7. Computational verification for $6 \leq n \leq 17$

For the values $n \in \{8, 10, 11, 12, 15\}$ not covered by Corollary 2 we verified existence by formulating the problem as a mixed-integer program: variables $x(S) \in \{0, 1\}$ for the winning status of each coalition S , the monotonicity constraints $x(S) \leq x(T)$ for $S \subseteq T$, the normalization $x(\emptyset) = 0$, $x(N) = 1$, and the swing equations $\sum_{S \not\ni i} (x(S \cup \{i\}) - x(S)) = 2c$ ($i \neq n$), $\sum_{S \not\ni n} (x(S \cup \{n\}) - x(S)) = c$, with $c \geq 1$ integer. Using HiGHS, all these instances are feasible. The table below summarizes the scale c found ($n = 6, 7$ by exhaustive enumeration; $n = 9, 13, 14, 16, 17$ by the explicit games of Section 5; the five remaining values by mixed-integer linear programming (MILP)). Each of those five values is reproduced at the same scale by the closed-form $\deg_G + r_i = D$ constructions of Corollary 16.

n	6	7	8	9	10	11	12	13	14	15	16	17
c	4	6	8	6	10	14	16	12	16	20	12	24

Note. The scale $c = 20$ for $n = 15$ was found by the CP-SAT model, an improvement over the previously reported $c = 22$; minimality at $n = 15$ is open

(a downward sweep at $c = 18$ did not terminate within a 600 s solver budget). Minimality is certified for $n = 8$ ($c = 8$, with $c \in \{2, 4, 6\}$ proven infeasible), $n = 9$ ($c = 6$, $c \in \{2, 4\}$ infeasible) and $n = 10$ ($c = 10$, $c \in \{2, 4, 6, 8\}$ infeasible); see Open problem 2 in Section 11.

For $n = 6$ the existence is established by exhaustive enumeration over all 7,828,352 simple games on six voters (360 solutions in two symmetry classes). For $n = 7$, exhaustive enumeration of all games is infeasible; we instead enumerated the 5-uniform family (winning coalitions of size 5) exhaustively and found 490 solutions, all of the complement-of-edges type $C_6 \cup star_7$. Thus Conjecture 16 holds for every $6 \leq n \leq 17$ by the constructions of Sections 3–5; the value $n = 18$ and all larger values are covered by the closed-form constructions of Section 11 (Corollaries 14–17).

8. Summary of the constructive half

For the inverse voting-design problem, we have delivered the *construction half* in full: an explicit, closed-form simple game whose Penrose–Banzhaf index equals $\psi^n = (2, \dots, 2, 1)/(2n - 1)$ for every $n \geq 6$, constructively resolving Kurz’s Conjecture 16, whose benchmark status motivated the design study. The main ingredients are:

1. **Two closed-form families** with exact swing formulas (Theorems 1 and 3): the *complement-of-edges* family (Theorem 1; covering $n \equiv 0, 2, 3, 6 \pmod{7}$) and the *threshold-with-exceptions* family (Theorem 3; supplying the small values 8, 10, 11, 12, 15), unified in a single parametric framework, the W-family (Proposition 11), whose swing formula (U) is exact and whose triangle instances (Proposition 13) cover the remaining classes $n \equiv 1, 4, 5 \pmod{7}$;
2. **A closed-form generalization to the whole target family** $\psi^n(a, b) \propto (a, \dots, a, b)$ (Theorem 2), which reduces realization by the complement-of-edges family to an explicit integrality condition and locates the covered sizes in residue classes modulo $3a+b$; the denser triangle route to the same family is compared with it in Section 9;
3. **A correction of the only published example** (Kurz and Napel 2014, footnote 19);
4. **A complete closed-form construction for every $n \geq 6$** (Corollaries 14, 16, 17), where the five smallest values 8, 10, 11, 12, 15 are handled by

the $\deg_G + r_i = D$ structure.

The arithmetic characterization (3) shows that no single complement-of-edges construction can cover all n , and the residue classes differ genuinely; nevertheless, a single parametric framework, the W -family with base degree and a suitable number of (special) triangles, realizes ψ^n for every n .

9. The weighted case: no weighted majority game realizes ψ^n for $6 \leq n \leq 10$

We now turn to the *obstruction half* of the design decision problem: is the target ψ^n realizable within the weighted majority class, the rule class used by most real voting bodies (parliaments, the EU Council, the IMF, corporate voting)? This is the question a designer must answer before deciding whether to stay in the weighted class. It is precisely Kurz’s **Conjecture 15**, the natural “dual” of Conjecture 16. As stated by Kurz (2015, Conjecture 15) it is quantitative: there exists a universal $c > 0$ with $\|Bz(v) - \psi^n\|_1 \geq c/n$ for every $n \geq 2$ and every weighted majority game v . In particular no weighted majority game reproduces ψ^n exactly (distance zero is excluded). We certify the exact case for $n \leq 10$, and the quantitative form for $n \leq 8$. Recall that a **weighted majority game** is a simple game $[q; w_1, \dots, w_n]$ with integer weights $1 \leq w_1 \leq \dots \leq w_n$ and quota $q \geq 1$, where S is winning iff $W(S) = \sum_{i \in S} w_i \geq q$. Let $\eta_i = \beta_i([q; w])$ be the swing count. The following identity is a classical structural fact for weighted majority games, via a swap/block argument of the type long used in the Banzhaf literature (cf. Felsenthal & Machover 1998). Applied to Kurz’s specific target ψ^n , the identity is the operative part of the obstruction: it pins the deviation on a subset-sum problem and yields the rigidity that the certification below exploits.

Proposition 4 (structural decomposition; swap/block argument). Let $v = [q; w]$ be a weighted majority game with $w_i \geq w_j$. Then $\eta_i - \eta_j = 2 \cdot |\{T \subseteq N \setminus \{i, j\} : q - w_i \leq W(T) < q - w_j\}|$. (K)

Proof. Enumerate the swings of i and j by the subset $T \subseteq N \setminus \{i, j\}$ that excludes both, and by whether the swing coalition S does or does not contain the other player: - i swings on $S = T$ iff $W(T) \in [q - w_i, q)$; - i swings on $S = T \cup \{j\}$ iff $W(T) \in [q - w_i - w_j, q - w_j)$; - j swings on $S = T$ iff $W(T) \in [q - w_j, q)$; - j swings on $S = T \cup \{i\}$ iff $W(T) \in [q - w_i - w_j, q - w_i)$.

Because $w_i \geq w_j$, we have $q - w_i \leq q - w_j$ and $q - w_i - w_j \leq q - w_i$, so $[q - w_j, q] \subseteq [q - w_i, q]$ and $[q - w_i - w_j, q - w_i] \subseteq [q - w_i - w_j, q - w_j]$, and in both inclusions the *removed* interval is the same one, $[q - w_i, q - w_j]$. Summing the four counts over all T and subtracting η_j from η_i cancels the two common pieces and leaves exactly twice the count of T with $W(T) \in [q - w_i, q - w_j]$. ■

The identity was verified on 12,000 random weighted games ($n = 3 \dots 12$, weights ≤ 30) with 0 failures.

Corollary 5 (parity; resolves Observation 2 in the Appendix). If $Bz(v) = \psi^n$, then the scale factor c is even. Two independent proofs: - (*any simple game*, Dubey-Shapley 1979): in every simple game all swing counts have a common parity. Here $\beta = (2c, \dots, 2c, c)$; since $2c$ is even, the common parity is even, so c is even. - (*weighted, from (K)*): if w_s is the special player ($\eta_s = c$), then for any $i \neq s$, $c = \eta_i - \eta_s = 2 \cdot |\{T : q - w_i \leq W(T) < q - w_s\}|$, so c is even.

Corollary 6 (the special player has strictly minimal weight). Suppose η_s is the unique minimum swing count and $\eta_i > \eta_s$ for every $i \neq s$. Then $w_s < w_i$ for all $i \neq s$. Indeed $\eta_i - \eta_s > 0$ forces the interval $[q - w_i, q - w_s)$ to contain some subset sum, so it is nonempty and $w_i > w_s$. Thus, in any weighted game whose Banzhaf index is proportional to ψ^n , the singled-out voter must be the *unique lightest* voter, an a-priori monotonicity constraint on the weights.

Corollary 7 (interval decomposition and the mirror obstruction). With s the special player and $\eta = (2c, \dots, 2c, c)$, identity (K) applied to the pair (i, s) yields, for every $i \neq s$, $|\{T \subseteq N \setminus \{i, s\} : q - w_i \leq W(T) < q - w_s\}| = c/2$, (L) so the excess $c/2$ of each non-special voter over the special voter is realized by subset sums of $N \setminus \{i, s\}$ inside the weight gap $[q - w_i, q - w_s)$. Now fix two non-special voters i, j with $w_i \geq w_j$ (WLOG) and split the window count in (L) at (i, s) according to whether j lies in T , and likewise the count at (j, s) according to whether i lies in T : $|\{T \subseteq N \setminus \{i, s\} : q - w_i \leq W(T) < q - w_s\}| = |\{T \subseteq N \setminus \{i, j, s\} : q - w_i \leq W(T) < q - w_s\}| + |\{T \subseteq N \setminus \{i, j, s\} : q - w_i - w_j \leq W(T) < q - w_s - w_j\}|$, $|\{T \subseteq N \setminus \{j, s\} : q - w_j \leq W(T) < q - w_s\}| = |\{T \subseteq N \setminus \{i, j, s\} : q - w_j \leq W(T) < q - w_s\}| + |\{T \subseteq N \setminus \{i, j, s\} : q - w_i - w_j \leq W(T) < q - w_s - w_i\}|$, both equal to $c/2$. Since $w_i \geq w_j > w_s$ (Corollary 6), the relevant weight windows nest as disjoint unions $[q - w_i, q - w_s) = [q - w_i, q - w_j) \sqcup [q - w_j, q - w_s)$, $[q - w_i - w_j, q - w_s - w_j) = [q - w_i - w_j, q - w_s - w_i) \sqcup [q - w_s - w_i, q - w_s - w_j)$,

so subtracting the two equalities leaves the **mirror identity** $|\{ T \subseteq N \setminus \{i, j, s\} : q - w_i \leq W(T) < q - w_j \}| + |\{ T \subseteq N \setminus \{i, j, s\} : q - w_s - w_i \leq W(T) < q - w_s - w_j \}| = 0$. (M) Both summands are nonnegative, hence they vanish individually: $|\{ T \subseteq N \setminus \{i, j, s\} : q - w_i \leq W(T) < q - w_j \}| = 0$ and $|\{ T \subseteq N \setminus \{i, j, s\} : q - w_s - w_i \leq W(T) < q - w_s - w_j \}| = 0$. (M') In words: for every pair of non-special voters i, j with $w_i \geq w_j$, the weight-gap window $[q - w_i, q - w_j)$, and its translate by w_s , is *sum-free* for the profile of the remaining $n - 3$ voters. Together with (L) this is a sharp rigidity statement: the $c/2$ sums realizing voter i 's excess over s must come exclusively from the part $[q - w_i - w_j, q - w_s - w_j)$ of the gap window, never from the sub-window $[q - w_i, q - w_j)$ itself; and the reflected sub-window $[q - w_s - w_i, q - w_s - w_j)$ is likewise empty. (Verified numerically: across 2.28×10^5 random weighted games with $\eta_i = \eta_j$, both windows in (M') were empty in every case.)

The Banzhaf-weighted obstruction property (M'). We record the rigidity statement just derived as the paper's central structural obstruction. Under the exact hypothesis $\eta = (2c, \dots, 2c, c)$, *any* realizing game must satisfy (L) and (M'); they are necessary conditions. They are also so tight that the full constraint system, (L), (M'), together with the exact swing profile, is *provably infeasible*, as certified by CP-SAT at the weight bound $w_i \leq 2^{n-1}$, which Muroga's bound validates on this range, for $n = 6, \dots, 10$ (Section 9, below). In this sense (M') is the structural certificate of the obstruction: the subset-sum windows it forces to be empty are precisely the degrees of freedom whose infeasibility the deterministic solver establishes. The property does computational work: the mirror constraints of (M') are the pruning constraints of the enhanced feasibility model, which decides $n = 6$ in 2.5 s (vs 2.9 s) and $n = 7$ in 19 s, about six times faster than the unpruned model at $n = 7$ (117 s). They are thus what makes the $n \geq 9$ frontier approachable at all. We are not aware of this reflected sum-freeness rigidity in the threshold-logic or voting-power literature. The closest structural studies, Chow's (1961) theorem on Chow parameters (a recovery problem) and the inverse-power-index certification results of Alon & Edelman (2010) and Kurz (2016), address reconstruction and worst-case approximation, not the exact-hypothesis rigidity used here.

Two-tier generalization $\psi^n(a, b)$ and a family-level no-go. Two routes from the single target to the family $\psi^n(a, b) = (a, \dots, a, b)/((n-1)a + b)$ are available, and they are complementary. The first is complement-of-edges: Theorem 2 reduces the existence of a realizing graph, for arbitrary $a > b \geq 1$, to

the integrality of $c = t(a-b)$ and $d = \lceil t(3a+b)/2 - 2(n-1) \rceil / (n-4)$; the share of sizes it reaches depends on the ratio, from two of thirteen residue classes at 4:1 to twelve of fifteen at 4:3, with four of seven at 2:1 (Section 3). The second, the triangle instances of the W-family (Proposition 13), is much denser in those cases where both have been computed: at the ratio 3:1 it covers every $n \leq 80$, where the complement-of-edges route covers four of ten classes; it realizes $\psi^n(a, b)$ with special degree $s = \lceil (a-b)(5n-5-p) - 4aq + 12b \rceil / (3a+b)$, (N) where p pure and q special triangles are used. Systematic search gives: - **(3,1) is uniform**: for all $n \in [6, 80]$, $s = n-1$ with $p = 0$, $q = 1$ realizes $\psi^n(3, 1)$, via the construction $G = \text{star}(n) \cup C_{n-1} \cup \{1, 2\}$ with F the single triangle on the special player. The same $p = 0$, $q = 1$, $s = n-1$ shape is used across the whole range, with no case distinction in n . - **(k,1), $k \geq 4$ is obstructed within this family**: for $k \geq 4$ the required degree $s \approx 5(k-1)n/(3k+1)$ exceeds $n-1$ for large n , so no triangle construction of this shape exists. Coverage rates over $n \in [6, 80]$: $k=2$: 73/75 (gaps 8, 11); $k=3$: 75/75; $k=4$: 40/75; $k=5$: 28/75; $k=6$: 9/75; $k=7$: 8/75. Corollaries 6–7 give the mechanism: Corollary 6 pins w_s minimal, after which (L) and (M) over-constrain the subset-sum profile. Both act inside this family, so what is obstructed here is the triangle construction, not the target; whether some other construction realizes $\psi^n(k, 1)$ for $k \geq 4$ is Open problem 5 of Section 11.

Computational obstruction (exact). For ψ^n itself, exact feasibility of a weighted realization with swing vector $(2c, \dots, 2c, c)$ was settled by CP-SAT. Infeasibility is certified at the weight bound $W = 2^{n-1}$, which the classical bound licenses: Muroga (1971, proof of Theorem 9.3.2.1) gives every weighted simple game an integer representation with $0 \leq w_i \leq \alpha_n$, where α_n is the largest determinant of an $n \times n$ 0–1 matrix (and $\alpha_n \leq (n+1)^{(n+1)/2}/2^n$ by Hadamard’s inequality), with $\alpha_n = 9, 32, 56, 144, 320$ for $n = 6, \dots, 10$, each at or below 2^{n-1} . Every swing count of the target is positive, so no voter can be a dummy and no realizing representation needs a zero weight, which is why the search is over $[1, W]$. Infeasibility at $W = 2^{n-1}$ is therefore conclusive for these n :

n	$W = 2^{n-1}$	model	verdict	runtime
6	32	standard	INFEASIBLE	2.9 s
7	64	standard	INFEASIBLE	117 s
8	128	+symmetry breaking	INFEASIBLE	316 s

n	$W = 2^{n-1}$	model	verdict	runtime
9	256	enhanced (feas2)	INFEASIBLE	1562.9 s
10	512	enhanced (feas2)	INFEASIBLE	9527.4 s

The bound is not uniform in n . The exact values α_n sit at or below 2^{n-1} through $n = 10$, but $\alpha_{11} = 1458$ already exceeds $2^{10} = 1024$ (at $n = 11$ Hadamard’s bound is attained). So $W = 2^{n-1}$ is a conclusive bound exactly on the range claimed here, $n \leq 10$, and the Tier-3 labelling of $n \geq 11$ is not a matter of solver budget alone. The values α_n are tabulated in the classical maximal-determinant problem (OEIS A003432); the current best integer-weight bound for weighted simple games is Kawana & Matsui (2022), which sharpens Muroga’s constants.

The pipeline is three-tiered, and we are explicit about which tier backs each claim. **Tier 1 (exhaustive)**: $n = 6$ is settled by enumerating all 1,111 weighted games, guaranteed. **Tier 2 (deterministic)**: $n = 7, 8, 9$ and 10 are settled by CP-SAT at the conclusive weight bound $W = 2^{n-1}$ (standard model for $n = 7, 8$; the enhanced model below for $n = 9, 10$), guaranteed. **Tier 3 (budgeted)**: $n \geq 11$ runs the same solver under a time budget, so a negative outcome there is heuristic evidence for a trend, not a proof, and is labeled accordingly.

$n = 6$ agrees with the exhaustive enumeration of all 1,111 weighted games on 6 voters whose best L_1 -distance to ψ^6 is $5/44 \approx 0.1136$; hence no weighted game attains distance 0. The $n = 8$ run required the symmetry-breaking constraint that non-special voters can be relabeled in nondecreasing weight order (all have equal target swing $2c$); with it the infeasibility certificate is found in 316 s, whereas the unconstrained model is inconclusive after 20 min. $n = 9$ is substantially harder: the standard model exhausts a full 2 h budget without a certificate (UNKNOWN), a dramatic jump consistent with the exponential growth of the subset-sum constraint structure. The enhanced model, which also enforces the sharp mirror constraints (M'), valid since they are consequences of the exact hypothesis, then decides $n = 9$ in 1562.9 s (3584 mirror constraints): **INFEASIBLE**. At $n = 10$ the same enhanced model decides **INFEASIBLE** in 9527.4 s (9216 mirror constraints), while the standard model again exhausts a 2 h budget without a certificate (UNKNOWN). This closes the exact obstruction for all $n \leq 10$, the strongest conclusive statement currently available; the enhanced model already gives a $6\times$ speedup on the smaller scales ($n = 6$: 2.5 s, $n = 7$: 19 s), and its growing mirror-constraint set is what makes the $n \geq 9$ frontier

approachable at all. This is exactly Kurz’s Conjecture 15, which asserts the quantitative statement that for some universal $c > 0$, $\|\text{Bz}(v) - \psi^n\|_1 \geq c/n$ for *every* weighted majority game v and every $n \geq 2$. The exact obstruction for $n = 6, \dots, 10$ is the first step; the constant c is extracted from the optimal L_1 errors computed just below.

Optimal L_1 -distance (exact computation). For each n we compute the exact minimal value of $\|\text{Bz}(v) - \psi^n\|_1$ over all weighted majority games, by fixing the total number of swings $T = \sum \eta_i$ and minimizing the integer objective $\sum_{i \neq s} |(2n-1)\eta_i - 2T| + |(2n-1)\eta_s - T|$ in CP-SAT, enumerating T over its full range and using the same weight bound $W = 2^{n-1}$; the special player is fixed at position n without loss (relabeling). The method reproduces the exhaustive $n = 6$ optimum exactly (for $n = 6$, $T \leq 45$): $L_1 = 5/44 \approx 0.113636$ at $T = 32$, **with weights $w = (3,1,3,3,3,1)$, quota $q = 11$** , matching the brute-force enumeration of all 1,111 weighted games, a strong check of both methods.

n	optimal L_1	T	$n \cdot L_1$	1/n
6	$5/44 \approx 0.113636$	32	≈ 0.682	0.167
7	≈ 0.085470	45	≈ 0.598	0.143
8	≈ 0.066667	60	≈ 0.533	0.125

(The $n = 8$ value is certified as the global optimum: $T = 60$ is OPTIMAL by CP-SAT, and the total swing count was swept over its full range to the theoretical maximum $8 \cdot 2^7 = 1024$, the full range sweep; its best value outside $T = 60$ is $L_1 = 0.116667$ at $T = 280$, well above the incumbent.) For $n = 7$ the optimal profile is $(5,7,7,7,7,5)$ with weights $(2,64,64,64,64,1)$ and quota 65; the value is certified OPTIMAL by CP-SAT (re-verified with a 600 s budget), and the full sweep over $T \leq 448$, the theoretical maximum $7 \cdot 2^6$ of the total swing count, confirms that no other total swing count improves on it: the best value in the complementary range $T \geq 121$ is $L_1 = 0.119535$ at $T = 139$, well above the incumbent. For $n = 8$ the optimum is attained at profile $(6,8,8,8,8,8,6)$ with weights $(2,3,3,3,3,3,1)$ and quota 4. The column $n \cdot L_1$ is the empirical anchor for the universal constant in Conjecture 15: since $\|\cdot\|_1 \geq c/n$, any admissible c is at most $\inf_n n \cdot L_1(n)$; the values above give $c \leq \min(30/44, 7 \cdot 0.08547, 8 \cdot 0.066667) \approx 0.533$. Each of the three values is an optimum over the entire range of the total swing count: $5/44$ for $n = 6$ follows from the exhaustive enumeration of all 1,111 weighted games, and the $n = 7$ and $n = 8$ values are

certified optimal with the total swing count swept to its theoretical maximum. The exact target $(2c, \dots, 2c, c)$ is infeasible for $6 \leq n \leq 10$ by the computation above, so the L_1 -*optimum* is strictly positive. It is the monotone decrease $n \cdot L_1 = 0.682 \rightarrow 0.598 \rightarrow 0.533$ that supports the *conjectured* $\Theta(1/n)$ asymptotics for all n ; deriving an analytic lower bound $\|Bz(v) - \psi^n\|_1 \geq c/n$ with an effective constant is open, and certification on the first three scales does not settle it. The optimal profiles are asymmetric: the two lightest voters, the special voter and one further voter, carry the reduced swing count, and they do not sit at the ideal scale $T = (2n - 1)c$ ($n = 6$: profile $(6, 4, 6, 6, 6, 4)$ at weights $(3, 1, 3, 3, 3, 1)$, quota 11; $n = 7$: profile $(5, 7, 7, 7, 7, 7, 5)$ at weights $(2, 6, 6, 6, 6, 6, 6, 1)$, quota 65; $n = 8$: profile $(6, 8, 8, 8, 8, 8, 8, 6)$ at weights $(2, 3, 3, 3, 3, 3, 3, 1)$, quota 4). This is consistent with Corollary 6: in the exact case the special voter alone would be uniquely lightest, whereas near-optimal games distribute the reduction between the two lightest voters.

10. Computational methodology: exact construction and certified obstruction

This section collects the two computational engines behind the paper as a reusable methodology, with the complexity statements and instance tables expected of a computational study. Every claim below is backed by a named script and a full log in the companion code package (Section 11).

10.1 Constructive engine: closed forms and the W-family MILP

Problem (existence). Given n , produce an explicit simple game with $Bz(v) = \psi^n$, or show that the family under consideration cannot cover n .

1. **Residue arithmetic.** For the complement-of-edges family, realization reduces (Theorem 1) to the diophantine equation $7c = d(2n - 8) + 4(n - 1)$ in positive integers d, c with $1 \leq d \leq n - 2$, $d + c/2 \leq n - 1$, together with Erdős–Gallai graphicality of the degree sequence $(d, \dots, d, d + c/2)$. This is an $O(1)$ decision per n , and the family covers exactly the four residue classes $n \equiv 0, 2, 3, 6 \pmod{7}$ (Proposition 8).
2. **W-family MILP/CP-SAT.** For the remaining residue classes, realization over the W-family of Proposition 11 is formulated as a MILP with edge variables x_e ($e \in G$) and triple variables y_T ($T \in F$), the bilinear products $z = y \cdot x$ linearized in the standard way. The exact swing for-

mula (U) and the monotonicity constraint $\text{Pairs}(F) \subseteq G$ enter as linear constraints. **Model size:** $O(n^3)$ **variables and constraints** ($C(n,2) + C(n,3)$ variables), versus 2^n for the general monotone Boolean MILP of Section 7, which in the benchmark of Section 10.4 returns a verdict only through $n = 10$. The compressed model solves every instance up to $n = 64$ in seconds to minutes (instance data below); its practical limiting factor is memory, reached around $n \approx 80$.

3. **Closed forms.** Proposition 13 gives explicit triangle constructions for the three hard residue classes $n \equiv 1, 4, 5 \pmod{7}$, valid for all $n \geq 22, 18, 19$ respectively; the small values 8, 10, 11, 12, 15 are handled by the $\deg_G + r_i = D$ structure (Corollary 16). Combined with Proposition 8 and Corollary 14, **every $n \geq 6$ admits an explicit closed form** (Corollary 17).

Representative W-family instances (hard residue classes; c = scale factor). All are explicit closed forms, each re-verified to have $\text{Bz}(v) = \psi^n$ by counting swings directly from the game definition rather than through (U) (full data in the companion package):

n	n mod 7	c	edges	triples	construction
18	4	24	33	6	$\deg+r = D = 4$
22	1	26	32	5	$\text{tri}(D3, f5, q3)$
25	4	30	36	6	$\text{tri}(D3, f6, q3)$
39	4	50	62	6	$\text{tri}(D3, f6, q2)$
67	4	90	114	6	$\text{tri}(D3, f6, q0)$
71	1	96	123	5	$\text{tri}(D3, f5, q0)$
74	4	100	127	6	$\text{tri}(D3, f6, q0)$
78	1	106	136	5	$\text{tri}(D3, f5, q0)$

Here $\text{tri}(D, f, q)$ denotes the triangle construction of Proposition 13 with base degree D , f disjoint triangles and q special triples (containing the special player), and $\deg+r = D$ the small-value structure of Corollary 16. The structural insight for the hard residues is that the solutions use few exceptional triples, $f \in \{4, 5, 6\}$, and that for $n \geq 67$ the special player need not participate in any special triple ($q = 0$; Corollary 15).

10.2 Verification engine: certification of weighted obstructions

Problem (decision). Given n , decide whether any weighted majority game $[q; w]$ on n voters realizes the exact target swing vector $(2c, \dots, 2c, c)$; if none exists, produce a certificate.

1. **Constraint system and conclusive weight bound.** The model has variables $w_1 \leq \dots \leq w_n \leq W$, q , and the swing counts η_i , together with the exact profile equations ($\eta_i = 2c$ for $i \neq s$, $\eta_s = c$) and the structural consequences (L), (M') of Section 9. Infeasibility at $W = 2^{n-1}$ is conclusive on the range claimed here: Muroga (1971, proof of Theorem 9.3.2.1) gives every weighted simple game an integer representation with $0 \leq w_i \leq \alpha_n$, where α_n is the largest determinant of an $n \times n$ 0–1 matrix (and $\alpha_n \leq (n+1)^{(n+1)/2}/2^n$ by Hadamard’s inequality), with $\alpha_n = 9, 32, 56, 144, 320$ for $n = 6, \dots, 10$, each at or below 2^{n-1} . The bound is not uniform in n ($\alpha_{11} = 1458$ already exceeds 2^{10}), which is why the certified claims stop at $n = 10$ and $n \geq 11$ is run budgeted and labelled Tier 3. A solver certificate of infeasibility at this bound is therefore a proof for that n , not a heuristic.
2. **Symmetry breaking.** Non-special voters may be relabeled in nondecreasing weight order (they carry equal target swing $2c$). Without this constraint $n = 8$ remains inconclusive after 20 min; with it the certificate is found in 316 s.
3. **Mirror pruning.** The sharp constraints (M') of Corollary 7 are consequences of the exact hypothesis, hence valid; enforcing them yields a roughly six-fold speedup ($n = 6$: 2.5 s vs 2.9 s; $n = 7$: 19 s vs 117 s, a six-fold speedup) and, decisively, makes the $n \geq 9$ frontier decidable at all.

Certified obstructions (deterministic, at the conclusive bound $W = 2^{n-1}$):

n	$W = 2^{n-1}$	model	verdict	runtime
6	32	standard	INFEASIBLE	2.9 s
7	64	standard	INFEASIBLE	117 s
8	128	+symmetry breaking	INFEASIBLE	316 s
9	256	+mirror	INFEASIBLE	1562.9 s
10	512	+mirror	INFEASIBLE	9527.4 s

The $n = 6$ verdict agrees with exhaustive enumeration of all 1,111 weighted games (Tier 1); $n = 7-10$ are Tier-2 deterministic certificates. Under a two-hour budget the standard (unpruned) model remains UNKNOWN at $n = 9$ and $n = 10$; the mirror constraints are what closes these cases. For $n \geq 11$ the same solver is run budgeted, so a negative outcome there is heuristic (Tier 3) and is labelled as such. The mirror constraints (3584 at $n = 9$ and 9216 at $n = 10$, *i.e.* $C(n-1, 2) \cdot 2^{n-2}$ window conditions, one per subset of the $n-3$ remaining voters and for both windows) are themselves a measure of the rigidity that (M') forces on any realizing game.

10.3 Exact optimization over total swing count

For the quantitative form of Conjecture 15 we compute the minimal L_1 -distance $\|Bz(v) - \psi^n\|_1$ exactly. Fixing the total number of swings $T = \sum \eta_i$, the integer objective $\sum_{i \neq s} |(2n-1)\eta_i - 2T| + |(2n-1)\eta_s - T|$ is minimized in CP-SAT; T is enumerated over its full range up to the theoretical maximum $n \cdot 2^{n-1}$, at the same conclusive weight bound. The method reproduces the exhaustive $n = 6$ optimum $5/44$ (weights $(3,1,3,3,3,1)$, quota 11) exactly, a strong cross-check of both methods. Certified optima over the full swing range: $n = 7$, $L_1 \approx 0.085470$ at $T = 45$; $n = 8$, $L_1 \approx 0.066667$ at $T = 60$ (over $T \leq 1024$). These give $c \leq \min(30/44, 7 \cdot 0.08547, 8 \cdot 0.066667) \approx 0.533$. At $n = 9$ the problem hits the CP-SAT complexity cliff: with 120 s per T -value, 23 of the 112 values in $[9,120]$ are decided, 0 are certified infeasible, and the best found $L_1 \approx 0.1042$ (at $T = 70$) is not certified. We report this honestly as the current frontier: **$c \leq 0.533$ is certified for $n \leq 8$; no certified value exists at $n = 9$.**

10.4 Benchmark: three formulations on one instance set

The engines above rest on a modelling decision, so we benchmark it. Three formulations of the same inverse problem were run against the same target ψ^n , on one machine, under a common 300 s solver time limit, over the instance set $n = 6, 7, 8, 9, 10, 12, 14, 16, 18, 20$.

F1, direct weight-space encoding ($2^n + n + 1$ variables). Variables are the weights $w_1 \leq \dots \leq w_n$, the quota q , the scale c , and one 0/1 indicator per nontrivial coalition, tied to the weighted-majority test by big-M rows, with the swing equations written directly over the indicators. This is the natural first attempt: it encodes the definition of Banzhaf power for a weighted game and

assumes no structure beyond that.

F2, general monotone Boolean game ($2^n + 1$ variables). The model of Section 7: one 0/1 variable per coalition, the monotonicity rows, and one swing row per voter, with no restriction to a graph class.

F3, compressed W-family model ($C(n, 2) + 4C(n, 3) + 1$ variables, hence $\Theta(n^3)$). The model of Section 10.1: edge variables on the $C(n, 2)$ pairs, triple variables on the $C(n, 3)$ triples, the bilinear products linearized in the standard way, and the exact swing formula (U) together with the monotonicity condition $\text{Pairs}(F) \subseteq G$.

n	F1: direct weighted	F2: general monotone	F3: compressed W-family
6	infeasible, 43.5 s	witness, 0.06 s	witness, 0.03 s
7	no verdict	witness, 0.12 s	witness, 0.05 s
8	no verdict	witness, 1.41 s	witness, 0.07 s
9	no verdict	witness, 0.75 s	witness, 0.07 s
10	no verdict	witness, 2.63 s	witness, 0.43 s
12	no verdict	no verdict	witness, 0.29 s
14	no verdict	no verdict	witness, 0.51 s
16	no verdict	no verdict	witness, 0.73 s
18	no verdict	no verdict	witness, 1.29 s
20	no verdict	build failed (memory)	witness, 2.14 s

Note. “Witness” means the solver returned a feasible 0/1 assignment, “infeasible” means it proved that no weighted majority game realizes the target, and “no verdict” means it returned neither within the 300 s limit. A measured wall time is given whenever a verdict was reached.

Model sizes, in variables, at five of the ten instance sizes:

n	6	10	14	18	20
F1	71	1035	16399	262163	1048597
F2	65	1025	16385	262145	—
F3	96	526	1548	3418	4751

The F2 entry at $n = 20$ is empty because the model could not be built; the sizes at the remaining instance sizes are in the companion package.

Three patterns are visible. The direct weighted encoding F1 is the least useful of the three: it settles $n = 6$, where it proves infeasibility in 43.5 s, and from $n = 7$ onward it spends the entire budget without producing any verdict. By $n = 20$ the model has grown to a million variables. The general monotone model F2, which drops the weight structure and searches all simple games, returns an explicit witness through $n = 10$ in at most 2.6 s, and then stops: no verdict at $n = 12, 14$ or 16 inside the budget, a 262,145-variable model built at $n = 18$ without a conclusion, and an out-of-memory failure while still building at $n = 20$. The compressed model F3 answers at every n in the set, from 0.03 s at $n = 6$ to 2.14 s at $n = 20$, and its sizes track the $\Theta(n^3)$ count exactly (96 variables at $n = 6$, 4,751 at $n = 20$). At $n = 18$ and $n = 20$, where F2 has stopped answering, F3 still answers in under 2.2 s. What the table compares is three encodings of one problem, not the difficulty of the problem itself.

Independent verification of the witnesses. Every witness returned by F2 or F3 was rebuilt from the solver assignment and its swing counts recomputed by enumerating all 2^n coalitions directly, without using the formulation that produced it. All 15 witnesses reproduce the target: the enumerated swing vector is $(2c, \dots, 2c, c)$, and divided by its common factor it is exactly $(2, \dots, 2, 1)$. For $n = 6$ through $n = 10$ both formulations returned a witness, and the normalized swing vectors agree.

Positive control. A benchmark that only ever returned negative answers would say nothing about the models. The same encoders were therefore run against a target that is known to be realizable by a weighted majority game, the unanimity vector $(1, \dots, 1)$, realized by equal weights with quota n . F1 returns that witness at $n = 6$ (71 variables) and at $n = 8$ (265 variables). The infeasibility it reports at $n = 6$ for ψ^n is therefore a property of the target and not an artefact of the encoding, and it corroborates the $n = 6$ case of Section 9 without using the structural conditions (L) and (M') that Section 9 imposes.

Limits of what the benchmark shows. Four qualifications belong with the table.

1. **No model carries an objective.** A reported scale c is therefore the scale of whichever feasible point the solver reached, not the minimal one. F3's scale coincides with the minimal c of the certified dataset in seven of the ten instances ($n = 6, 7, 8, 9, 10, 14, 16$) and exceeds it in the other

three (24 against 16 at $n = 12$; 28 against 24 at $n = 18$; 58 against 20 at $n = 20$). The table measures the time to produce a realizing game. Minimality is settled separately, by the closed forms of Section 10.1 and the certified dataset, and it is not what the timings compare.

2. **F1 is our own reconstruction of the natural direct encoding.** It represents the formulation a first attempt would write. It is not a third-party implementation, and nothing above should be read as a comparison against the solvers or runtimes of any published treatment of the inverse power index problem, including the computational verifications of Kurz and Napel (2014) and de Keijzer et al. (2014), which use different formulations on different instance sets.
3. **The solver does not always stop when told to.** Several rows were given a 300 s limit and ran longer; the times reported here are measured wall times, and the largest overshoot is $2.18\times$ (F2 at $n = 12$).
4. **The frontier is measured, not proved.** One machine and ten instance sizes give an empirical ranking. Only the size column follows from the models themselves: $2^n + n + 1$, $2^n + 1$, and $C(n, 2) + 4C(n, 3) + 1$ variables respectively.

Four points about the table deserve a word on their relation to earlier sections. The MILP of Section 7 is the same model as F2, differing only in the bound placed on c , and its role was to discover the instances $n \in \{8, 10, 11, 12, 15\}$ before the closed forms of Corollary 16 were available. Its $n = 12$ instance is one that the common 300 s cap does not resolve, while existence of a game at the recorded scale $c = 16$ is today certified by the closed form of Corollary 16, so the timed row takes nothing away from the existence result. The $n = 6$ infeasibility of F1 is established at the same conclusive weight bound $W = 2^{n-1} = 32$ that Section 9 uses ($\alpha_6 = 9 \leq 32$), and F1 imposes neither (L) nor (M'), so it certifies that case without recourse to the structural conditions. The sizes of the second table are the $2^n + n + 1$, $2^n + 1$ and $C(n, 2) + 4C(n, 3) + 1$ counts of Proposition 12, read off the models as they were built. Finally, the F2 column records what the shared cap reached, not where the model stops. Re-run outside the benchmark with a per-instance budget, the Section 7 copy returns a witness at $n = 11$ in 25.9 s, and at $n = 12$ in 270 s under its looser bound $c \leq 100000$; the benchmark's own copy, with $c \leq 100$, reaches the same scale $c = 20$ at $n = 12$, again not the minimal one, but needs 919 s to do it. Solver time on this model is sensitive to the bound placed on c , which does not

change which games exist.

10.5 Reproducibility

Every numerical claim above is reproduced by the companion code package, which ships one script per claim together with its raw solver logs. The per-claim script inventory, the solver versions, the hardware, and the per-run parameters are listed in the package README; the manuscript itself names no package file, so that the argument can be read without the package at hand.

11. The W-family: unified constructions, structural properties, and open problems

This section develops the W-family into its full scope: it proves the unified swing formula that the construction half (Section 8) relies on, establishes the structural properties that govern which residue classes admit closed-form realizations, and collects the open problems that remain. All results are labelled with their derivation or verification method; unproven numerical observations are explicitly marked as such. Reproducible code is given in the appendix.

Proposition 8 (Exact description of complement-of-edges coverage; Erdős–Gallai computation)

For each $n \geq 6$, the complement-of-edges family (Theorem 1) is solvable at n if and only if there are positive integers d, c with $7c = d(2n - 8) + 4(n - 1)$, $1 \leq d \leq n - 2$, c even, $d + c/2 \leq n - 1$, $(n \cdot d + c/2)$ even, and the degree sequence $(d, \dots, d, d + c/2)$ passes the Erdős–Gallai test and is constructible by Havel–Hakimi.

Lemma A (two-value Erdős–Gallai reduction). Let $1 \leq d < D \leq n - 1$. The degree sequence $(d, \dots, d, D)$ with $n - 1$ copies of d is graphical if and only if $(n - 1)d + D$ is even, $D \leq n - 1$, and $D \leq d(n - d)$.

Proof. Sorting descending as $(D, d, \dots, d)$, the Erdős–Gallai inequalities are, for every k , $\sum_{i \leq k} a_i \leq k(k - 1) + \sum_{i > k} \min(a_i, k)$. For $k = 1$ this reads $D \leq n - 1$. For $2 \leq k \leq d$ every $a_i = d \geq k$, so the RHS is $k(k - 1) + (n - k)k = k(n - 1)$; the strongest such inequality is at $k = 2$, i.e. $D \leq 2(n - 1) - d$, which is implied by $D \leq n - 1$. For $k > d$ the RHS is $k(k - 1) + (n - k)d$, and the strongest case is $k = d + 1$, reading $D \leq d(d + 1) + (n - d - 1)d = d(n - d)$. Hence,

besides parity and $D \leq n - 1$, the only content is $D \leq d(n - d)$. (Sufficiency is the Erdős–Gallai theorem; Erdős & Gallai 1960.) ■

Analytic conclusion. Applying Lemma A to the four residue classes with the constant d above ($n \equiv 6, 0, 2, 3 \pmod{7}$; $D = d + c/2 = (4n+4)/7, (5n+7)/7, (3n+1)/7, (6n+10)/7$), the conditions hold for every admissible n : for $n \equiv 6 \pmod{7}$ ($d = 2$) both $D \leq n - 1$ and $D \leq 2(n - 2)$ hold for all $n \geq 6$; for $n \equiv 0 \pmod{7}$ ($d = 3$) $D \leq 3(n - 3)$ always and $D \leq n - 1$ for $n \geq 7$; for $n \equiv 2 \pmod{7}$ ($d = 1$) both hold for all $n \geq 9$; for $n \equiv 3 \pmod{7}$ ($d = 4$) $D \leq 4(n - 4)$ always and $D \leq n - 1$ for $n \geq 17$. Parity: $(n - 1)d + D$ equals $(18m+14), (26m - 2), (10m+2), (34m+12)$ respectively for $n = 7m+r$, all even. Hence the complement-of-edges family covers $n \equiv 6 \pmod{7}$ from $n \geq 6$, $n \equiv 0 \pmod{7}$ from $n \geq 7$, $n \equiv 2 \pmod{7}$ from $n \geq 9$, and $n \equiv 3 \pmod{7}$ from $n \geq 17$, with no computational verification. (The value $n = 10$, the only $n \equiv 3 \pmod{7}$ with $6 \leq n < 17$, is covered by Corollary 16.)

Corollary 9 (new infinite families). All admissible n in the following four residue classes are realized by complement-of-edges constructions: $n \equiv 6 \pmod{7}$, $d = 2$; $n \equiv 0 \pmod{7}$, $d = 3$; $n \equiv 2 \pmod{7}$, $d = 1$; $n \equiv 3 \pmod{7}$, $d = 4$. Each construction is a complement-of-edges game whose graph is “a star centered at the special player plus a structured graph”; explicit graphs for $n = 20, 21, 23, 24$ etc. were built and independently re-checked by both the swing formula and (for small n) exhaustive enumeration. The complement-of-edges family therefore covers **4/7 of all integers**, far beyond the two residue classes of the main text.

Uncovered residue classes: $n \equiv 1, 4, 5 \pmod{7}$ ($n = 18, 19, 22, 25, 26, \dots$ in the main text).

A structural property used as a search restriction

Observation 1 (certified for $n \leq 19$; not used as a hypothesis in any proof). For every simple game v on $n \leq 19$ voters with $Bz(v) = \psi^n$, every $(n - 1)$ -element coalition is winning.

Evidence: a losing $(n - 1)$ -set falls into one of two cases, because the symmetry group of the problem permutes the voters whose Banzhaf coefficients agree: the set omits a non-special voter, or it omits the special one, and within the first case all choices are equivalent. A model with one Boolean per coalition, ranging over every monotone Boolean function rather than only the weighted games, is infeasible in both cases for every $n \leq 19$, in integer arithmetic and with no time

limit. For $n = 6$ the same conclusion follows from the exhaustive enumeration of all 7,828,352 simple games on six voters, whose 360 solutions satisfy it. The same model without the losing-set constraint is feasible, and the runs return infeasible with the symmetry break removed as well, so the verdict depends neither on the encoding nor on the break.

Why it is plausible: if $N \setminus \{x\}$ is losing, player x swings on $S = N \setminus \{x\}$; moreover, on each $(n - 2)$ -set $S = N \setminus \{x, j\}$, player x may swing (because $N \setminus \{j\}$ is winning) while player j never does (because $N \setminus \{x\}$ is losing), so x accumulates extra swings at the top levels, contradicting $\beta \propto (2, \dots, 2, 1)$. What remains open is the passage from the certified range to every n : a proof would need a level-by-level count of the kind used in Section 11 (Problem 1 below).

Where it is used: only as a restriction on the search for constructions: we require all $(n - 1)$ -sets to win when solving for (G, F) . This is sound for the existence statements that follow, because restricting a search can lose solutions but cannot invent them, and every construction reported here is independently re-verified by direct enumeration of its swing counts. The obstruction certificates of Section 9 do not use this property at all: that model constrains the weights and the quota directly.

A unified framework: the W-family

Proposition 11 (unifying theorem). For a graph G and a family F of triples with $\text{Pairs}(F) \subseteq G$ (monotonicity), define $v(G, F)(S) = 1 \iff |S| \geq n - 1$, or $(|S| = n - 2 \text{ and } N \setminus S \in G)$, or $(|S| = n - 3 \text{ and } N \setminus S \in F)$. Then $v(G, F)$ is a simple game and $\beta_i(v(G, F)) = |G| + (n - 1) + |F| - 2 \cdot \deg_G(i) - 2 \cdot r_i + e_i$, (U) where $r_i = |\{X \in F : i \in X\}|$ and $e_i = |\{X \in F : i \in X, X \setminus \{i\} \notin G\}|$.

Proof. Monotonicity first. Suppose $S \subseteq T$ and S is winning. If $|S| \geq n - 1$ then $|T| \geq n - 1$ and T wins. If $|S| = n - 2$ with $N \setminus S \in G$, then either $|T| \geq n - 1$, or $|T| = n - 2$ and $T = S$. If $|S| = n - 3$ with $N \setminus S \in F$, then either $|T| \geq n - 1$, or $|T| = n - 2$, and in that case $N \setminus T$ is a pair inside a triple of F , hence an edge of G by $\text{Pairs}(F) \subseteq G$, so T wins. Hence $v(G, F)$ is monotone, with $v(\emptyset) = 0$ and $v(N) = 1$: a simple game.

None of the four cases below uses $\text{Pairs}(F) \subseteq G$ either, so (U) is an identity for every (G, F) ; the hypothesis is needed only for the monotonicity just shown.

Now enumerate the losing S with $i \notin S$ and $S \cup \{i\}$ winning, by $|S|$. The case $|S| = n - 1$ cannot occur, since $N \setminus \{i\}$ is winning by the first clause of the definition.

- $|S| = n - 2$. Here $S \cup \{i\}$ has size $n - 1$ and is winning, so i swings exactly when S is losing. Since $N \setminus S = \{i, j\}$ for some $j \neq i$, that condition is $\{i, j\} \notin G$; each such j contributes one swing, giving $(n - 1) - \deg_G(i)$ swings.
- $|S| = n - 3$. Here $S \cup \{i\} = N \setminus \{a, b\}$ is winning iff $\{a, b\} \in G$, and $S = N \setminus \{i, a, b\}$ is losing iff $\{i, a, b\} \notin F$. The pairs $\{a, b\} \in G$ with $i \notin \{a, b\}$ number $|G| - \deg_G(i)$; from these we must discard those whose third vertex completes a triple of F . Of the r_i triples of F containing i , exactly e_i leave a pair outside G , so exactly $r_i - e_i$ of them lie over a pair of G . This gives $|G| - \deg_G(i) - r_i + e_i$ swings.
- $|S| = n - 4$. Here $S \cup \{i\} = N \setminus X$ for a triple X with $i \notin X$, and S is losing because $|S| < n - 2$; $S \cup \{i\}$ wins iff $X \in F$, giving $|F| - r_i$ swings.
- $|S| \leq n - 5$. Then $S \cup \{i\}$ has size at most $n - 4$ and is losing, so it yields no swing.

These cases exhaust the possible sizes of S , and summing the first three yields (U). ■

Verification. The check is case by case rather than on the total alone: on random (G, F) instances, each of the three partial counts matches the closed form it was derived with, and the three add to the brute-force swing count, over 2040 player-instances for $n = 6..11$ with $\text{Pairs}(F) \subseteq G$ and a further 2040 without it.

- With $G = E$ and $F = \emptyset$, (U) reduces to Theorem 1 (complement-of-edges);
- With $G = \text{complement}(E)$ and arbitrary F , (U) reduces to Theorem 3 (threshold-with-exceptions).

This unification shows that the two construction families belong to one parametric framework, and provides a single equation for finding new constructions.

Proposition 12 (an $O(n^3)$ -variable MILP over the W-family; verification extended to $n \geq 18$)

The general MILP of Section 7 has $2^n + 1$ variables; the benchmark of Section 10.4 obtains a verdict from it through $n = 10$, builds a 262,145-variable model

at $n = 18$ without concluding, and exhausts memory while building at $n = 20$. Using Proposition 11's W -family, we compress the search to variables x_e (edge $e \in G$, $C(n,2)$ of them) and y_T (triple $T \in F$, $C(n,3)$ of them), with the bilinear terms $z = y \cdot x$ linearized in the standard way, yielding a MILP with $O(n^3)$ **variables** (formula (U) and monotonicity $\text{Pairs}(F) \subseteq G$ as linear constraints). This MILP solves all instances up to $n \leq 64$ in seconds to minutes; every resulting construction is an explicit closed form (Corollaries 14–17, Proposition 13), re-verified by formula (U). The scales c are:

n	$n \bmod 7$	c	construction
18	4	24	deg+r=D
19	5	22	tri(D3,f4,q2)
20	6	20	comp-edge
21	0	26	comp-edge
22	1	26	tri(D3,f5,q3)
23	2	18	comp-edge
24	3	36	comp-edge

n	$n \bmod 7$	c	construction
25	4	30	tri(D3,f6,q3)
26	5	32	tri(D3,f4,q2)
27	6	28	comp-edge
28	0	36	comp-edge
29	1	36	tri(D3,f5,q2)
30	2	24	comp-edge
31	3	48	comp-edge

Here $\text{tri}(D, f, q)$ and **comp-edge** are as in Proposition 13 / Section 10.1.

Then $n = 32 (\equiv 4) c=40$, $n=33 (\equiv 5) c=42$, $n=34 (\equiv 6) c=36$, $n=35 (\equiv 0) c=46$, $n=36 (\equiv 1) c=46$; $n=37 (\equiv 2) c=30$ and $n=38 (\equiv 3) c=60$ by complement-of-edges; $n=39 (\equiv 4) c=50$. Further: $40(\equiv 5) c=52$, $43(\equiv 1) c=56$, $46(\equiv 4) c=60$, $47(\equiv 5) c=62$, $50(\equiv 1) c=66$, $53(\equiv 4) c=70$, $54(\equiv 5) c=72$, $57(\equiv 1) c=76$, $60(\equiv 4) c=80$, $61(\equiv 5) c=82$, $64(\equiv 1) c=86$, all rechecked by (U).

Closed forms cover every $n \geq 67$ (supplement to Proposition 12).

Corollary 15 shows that for $n \geq 67$ the special player need not participate in

any F-triple ($q = 0$). Applying the triangle construction directly yields the remaining values: $n = 67 (\equiv 4) \ c = 90$, $n = 68 (\equiv 5) \ c = 92$, $n = 71 (\equiv 1) \ c = 96$, $n = 74 (\equiv 4) \ c = 100$, $n = 75 (\equiv 5) \ c = 102$, $n = 78 (\equiv 1) \ c = 106$, with $|G| = 123$, $|F| = 5$ at $n = 71$ and $|G| = 136$, $|F| = 5$ at $n = 78$. **Hence every n with $6 \leq n \leq 80$ is covered without gaps, for all residue classes mod 7** (verified: 75/75 values in $[6, 80]$). The structural insight for the hard residues is that their solutions use few exceptional triples: the triangle construction needs at most six ($|F| \in \{4, 5, 6\}$), and for $n \geq 67$ none of them involves the special player. For $n \geq 81$ the $O(n^3)$ model exhausts memory around $n \approx 80$, recorded as the second computational lower bound of this method.

Methodological remark: the speed of the W-family MILP/CP-SAT ($n = 18$ in about 5 s) shows that the unified framework of Proposition 11 yields a scalable computational route to the constructions of this section.

Closed-form constructions for the hard residue classes; full resolution

Proposition 13 (found by exploration, verified via the proven formula (U)). For each of the three residue classes $n \equiv 1, 4, 5 \pmod{7}$, there is an explicit closed-form construction realizing ψ^n , valid for all n from a small threshold:

residue	f	c	special degree s	valid for
$n \equiv 1$	5	$(10(n-1) - 28)/7$	$(c+6)/2 - q$	$n \geq 22$
$n \equiv 4$	6	$(10(n-1) - 30)/7$ for $n \geq 25$ ($D = 3$); $c = 24$ at $n = 18$ ($D = 4$)	$D + c/2 - q$	$n \geq 18$ ($D = 4$ at $n = 18$, else $D = 3$)
$n \equiv 5$	4	$(10(n-1) - 26)/7$	$(c+6)/2 - q$	$n \geq 19$

Here $q \in \{0, \dots, f\}$ is the number of *special triples*, the F-triples containing the special player (equal to the number of triples in which the special player participates). The construction takes f disjoint triangles: q of them of the form $\{n, a, b\}$ (special triple) and $p = f - q$ of them involving only non-special players; every player in a triangle has degree 2 in G (base degree $D = 3$, so each triangle

player has degree $D - 1 = 2$) and lies in exactly one F-triple ($r_i = 1, e_i = 0$). The remaining $n - 1 - 3f + q$ non-special players have degree $D = 3$ and $r_i = 0$; the special player has degree $s = D + c/2 - q$ (for $D = 3$ this is $(c+6)/2 - q$). By formula (U) the swing counts are $\beta_i = |G| + (n - 1) + f - 6 = 2c$ (non-special), $\beta_n = |G| + (n - 1) + f - 2s - 2q = c$. The smallest feasible q admits the closed form $q = \max(0, \lceil (K - n)/14 \rceil)$, $K = 57$ ($n \equiv 1$), 67 ($n \equiv 4$), 47 ($n \equiv 5$). (6) Indeed, writing $\delta = 2, 1, 3$ for $n \equiv 1, 4, 5$ respectively, direct substitution ($D = 3, c = (10(n - 1) - A)/7$ with $A = 28, 30, 26$) gives the special degree $s = (5n + \delta)/7 - q$ and the residual degree sequence $(3^a, 2^x)$ on the $m = n - 1 - 3f + q$ non-triangle players, where $x = s - 2q$ and $a = m - x = (2(n - K) + 28q)/7$. Since $n \equiv K \pmod{7}$, a is **always even**. The feasibility conditions $x \geq 0, a \geq 0, a \leq (m+1)/2, q \leq f, m \geq 4, s \leq n - 1$ all hold at (6): $x \geq 0 \iff q \leq (5n + \delta)/21$ and $a \leq (m+1)/2 \iff 49q \leq 3n + 4K - 21f$ are the binding ones, each holding for every $n \geq n_0$ once the elementary inequalities are checked (e.g. $n \geq (3K + 42 - 2\delta)/13 < n_0$ for $x \geq 0$; $n \geq (3.5K + 49 - (4K - 21f))/6.5 < n_0$ for the chord condition); $a \geq 0$ is built into (6), $q \leq f$ and $s \leq n - 1$ are automatic. The residual graph $(3^a, 2^x)$ is then constructed explicitly: place the a degree-3 vertices at the even positions $0, 2, \dots, 2a - 2$, add the cycle $(0 \ 1 \ \dots \ m - 1 \ 0)$, and add the matching chords $(2j, 2j+2)$ for $j = 0, 2, \dots, a - 2$. Each even-position vertex is the endpoint of exactly one chord, so it has degree 3; all other vertices have degree 2. ($a \leq (m+1)/2 \iff 2a - 2 \leq m - 1$ makes the positions exist; a even closes the chord set.) Thus (6) realizes ψ^n for every $n \equiv 1, 4, 5 \pmod{7}$ with $n \geq 22, 25, 19$ respectively, with no search and no reliance on numerical verification.

Corollary 14 (every $n \geq 18$ has a closed form). Since $n \equiv 0, 2, 3, 6 \pmod{7}$ are covered by the complement-of-edges construction of Proposition 8 (by Lemma A, analytically for every admissible n) and $n \equiv 1, 4, 5$ by the general triangle construction of Proposition 13 (by the closed form (6), analytically for $n \geq 22, 25, 19$ respectively), **every $n \geq 18$ admits an explicit closed-form simple game with $\mathbf{Bz}(\mathbf{v}) = \psi^n$** . For $n = 18$ this uses base degree $D = 4$ with six disjoint triangles, three of which contain the special player: $c = 24, |G| = 33, |F| = 6$.

Corollary 15 (unified parametric family). With base degree $d = 3$ for all non-special players and $f =$ the smallest non-negative integer with $f \equiv 5(n \pmod{7}) \pmod{7}$, the same construction realizes ψ^n for every residue class; for $n \geq 67$ the special player need not participate in any F-triple, and for smaller n a

suitable number q of special triples is used. Thus a single parametric family (the W-family of Proposition 11 with the degree/f-triple data above) covers all n .

Corollary 16 (closed forms for the five remaining small values). For $n = 8, 10, 11, 12, 15$ explicit closed-form constructions are also found (via the same $\deg_G + r_i = D$ structure, with the data below; verified by formula (U)):

n	$n \bmod 7$	c	$ G $	$ F $	$D = \deg + r$ (non-special)
8	1	8	15	2	4
10	3	10	16	1	3
11	4	14	22	6	5
12	5	16	25	6	5
15	1	20	32	2	4

Each is a W-family game $v(G, F)$ with all swing counts $\propto (2, \dots, 2, 1)$ and $\text{Pairs}(F) \subseteq G$ (monotone), so $\text{Bz}(v) = \psi^n$.

Corollary 17 (full resolution, analytic). Combining Corollary 14 (analytic, via Lemma A and (6)) with Corollary 16, **every $n \geq 6$ admits an explicit closed-form simple game with $\text{Bz}(v) = \psi^n$** . The proof is fully analytic: for $n \equiv 0, 2, 3, 6 \pmod{7}$ the complement-of-edges construction of Proposition 8 realizes ψ^n for every admissible n (Lemma A: the two-value Erdős–Gallai conditions hold for all n in the residue class), and for $n \equiv 1, 4, 5 \pmod{7}$ the triangle construction of Proposition 13 with the closed form (6) realizes ψ^n for every $n \geq 22, 25, 19$ respectively; the finitely many remaining values $6 \leq n \leq 21$ are the explicit games of Corollary 16 ($n = 8, 10, 11, 12, 15$) and of Proposition 8 ($n = 6, 7, 9, 13, 14, 16, 17$), with $n = 18$ the $D = 4$ variant of Corollary 14. Kurz’s Conjecture 16 is therefore fully resolved with explicit constructions for all n , with no reliance on computer search for existence.

Verification of the core theorems (independent re-check)

The three central claims were re-verified independently of their original derivations:

(a) The W-family swing formula (U), Proposition 11. The identity $\beta_i(v(G, F)) = |G| + (n-1) + |F| - 2 \cdot \deg_G(i) - 2 \cdot r_i + e_i$ is proved in Section

11 by the same enumeration of $|S|$ that proves Theorem 3. Independently of the proof, the count was checked against direct swing enumeration on 440 randomly generated monotone instances ($\text{Pairs}(F) \subseteq G$, $n = 4..14$): **0 mismatches**; and the proof's case decomposition was checked on its own, each of the three partial counts against the closed form it was derived with and the three against the brute-force total, over 2670 player-instances for $n = 6..18$, and again over 2670 more with $\text{Pairs}(F) \subseteq G$ not imposed, (U) being an identity for every (G, F) : **0 mismatches** in either pass. (Forty random instances per n up to $n = 11$ and six per n above; the larger ones are brute-forced over up to 2^{18} coalitions.) The two special cases were checked separately, Theorem 1 ($F = \emptyset$, 199 random instances) and Theorem 3 ($G = \text{complement}(E)$, 29 instances), also with **0 mismatches**.

(b) The closed-form constructions of Proposition 13 (analytic by Lemma A and (6)). As an independent confirmation of the analytic proof, the triangle-with-special-triples construction was generated and its swing counts verified via (U) for every $n \in [6, 50]$ with $n \equiv 1, 4, 5 \pmod{7}$ and $n \geq 18$, and for the complement-of-edges constructions for the other residue classes: **all verified** (the five values 8, 10, 11, 12, 15 are covered by the $\deg_G + r_i = D$ constructions of Corollary 16, also verified). The Erdős–Gallai conditions of Lemma A and the feasibility inequalities of (6) were also checked by direct computation for all n up to 10^6 , agreeing with the analytic conclusion. The appendix constructions themselves were checked as a chain rather than through the formula: for every $n = 6..80$ the shipped data satisfy $\text{Pairs}(F) \subseteq G$ and the degree condition $\deg_G(i) + r_i = D$ with the special player $c/2$ above it, whereupon (U) returns $c \cdot (2, \dots, 2, 1)$; and for $n \leq 24$ the swing counts were enumerated directly from the game definition, with no passage through (U) at all.

(c) The search restriction of Observation 1. Exhaustive enumeration for $n = 6$ (all 360 solutions), and an exact decision of the two inequivalent cases of a losing $(n - 1)$ -set in integer arithmetic with no time limit, confirm that all $(n - 1)$ -sets are winning in any game with $\text{Bz}(v) = \psi^n$, for every $n \leq 19$; the same model is feasible once that constraint is dropped, which is the control that the infeasibility is not an artifact of the encoding.

All verification scripts are included in the reproducible code package.

Current coverage table

n	method	c
6	compl	4
7	compl	6
8	deg+r	8
9	compl	6
10	deg+r	10
11	deg+r	14
12	deg+r	16
13	compl	12
14	compl	16
15	deg+r	20
16	compl	12
17	compl	24

(18–80 see Proposition 12.) $n \equiv 0, 2, 3, 6 \pmod{7}$ are covered for infinitely many n by Proposition 8; $n \equiv 1, 4, 5 \pmod{7}$ are covered by the triangle construction of Proposition 13 and by Corollary 16 for the smallest values. **Every $n \geq 6$ has a closed-form construction.** Full explicit (G, F) data for $6 \leq n \leq 80$ are provided in machine-readable form in the companion data file; entries for $41 \leq n \leq 80$ are produced by the closed-form constructions of Proposition 8 ($n \equiv 0, 2, 3, 6 \pmod{7}$) and Proposition 13 ($n \equiv 1, 4, 5 \pmod{7}$), and every entry ($6 \leq n \leq 80$) is re-verified by direct enumeration of the swing counts (equal to ψ^n , i.e. $(2c, \dots, 2c, c)$) and of the monotonicity condition $\text{Pairs}(F) \subseteq G$.

Open problems

1. Can Observation 1 (all $(n-1)$ -sets winning) be proved rigorously? It is certified exactly for $n \leq 19$, so what the question asks for is the general n .
2. What is the minimal c for each n ? (Certified by the CP-SAT model of Section 7: $n=6$: 4, $n=7$: 6, $n=8$: 8, $n=9$: 6, $n=10$: 10.) For $n = 8$ the previously open “7 or 8” gap is closed: the infeasibility of every smaller even scale $c \in \{2, 4, 6\}$ is certified, and $c = 8$ is realized (the table of Section 7); likewise $n = 9$ ($c \in \{2, 4\}$ infeasible, $c = 6$ realized) and $n = 10$ ($c \in \{2, 4, 6, 8\}$ infeasible, $c = 10$ realized). Since c is even (Corollary 5), the odd candidate $c = 7$ is excluded; the certified values are conclusive because CP-SAT proves infeasibility exactly.

3. Is there a single uniform construction that realizes ψ^n for all n with no case distinction? The current resolution uses residue-specific parameters within one framework (Corollary 15).
4. (*Weighted case, Conjecture 15.*) Can the certified infeasibility of the exact target $(2c, \dots, 2c, c)$ for $6 \leq n \leq 10$ (Section 9) be turned into an analytic proof for all n ? In particular, the quantitative form asks for a universal $c > 0$ with $\|Bz(v) - \psi^n\|_1 \geq c/n$ for every weighted majority game v . The certified optimal distances $5/44$, 0.085470 , 0.066667 ($n = 6, 7, 8$) give the empirical constant $c \leq 0.533$; an analytic argument with $c = \Theta(1)$ would settle the conjecture.
5. (*W-family, the $(k, 1)$ targets for $k \geq 4$.*) Proposition 13 obstructs every triangle construction of the W-family at these targets, and Corollaries 6–7 explain why; both are statements about that family. Is $\psi^n(k, 1)$ for $k \geq 4$ weighted-realizable by some construction outside it? In the other direction, is there an argument confined to that family that no such construction can exist? The coverage of the family falls off with k (40, 28, 9 and 8 of the 75 values of n at $k = 4, 5, 6, 7$), so a positive answer would most likely be a uniform construction of a different shape.

12. Application: an exact computational case study

We close with a concrete case study applying the two engines of Sections 10.1–10.3 to a real voting body: the weighted voting rules of the Council of the European Union (EU) and its predecessors. Every claim below is tagged with its epistemic status: **certified** (deterministic computation at a conclusive weight bound), **exact** (closed form / subset enumeration), or **contextual** (institutional background, not a theorem of this paper).

12.1 Actual Banzhaf power of historical EU Council bodies

The original European Economic Community (EEC) Council (Treaty of Rome, 1958) and its first enlargements were weighted majority games on $n = 6, 9, 10$ voters, exactly the certified range of Section 9. Their official qualified-majority weights and quotas (Felsenthal & Machover 1998, who tabulate the EU Council history; the accession treaties) and the exact Banzhaf power distributions are:

body	n	weights w	quota	
			q	L_1 to ψ^n
EEC-6 (1958–1973)	6	$(4 \times 3, 2 \times 2, 1)$	12	$26/77 \approx 0.338$
EEC-9 (1973–1981)	9	$(10 \times 4, 5 \times 2, 3 \times 2, 2)$	41	$2136/5389 \approx 0.396$
EEC-10 (1981–1986)	10	$(10 \times 4, 5 \times 3, 3 \times 2, 2)$	45	$2528/6023 \approx 0.420$
EEC-12 (1986–1995) †	12	$(10 \times 4, 8, 5 \times 4, 3 \times 2, 2)$	54	$878/2323 \approx 0.378$
EU-15 (1995–2004) †	15	$(10 \times 4, 8, 5 \times 4, 4 \times 2, 3 \times 3, 2)$	62	$186466/480385 \approx 0.388$

Weights are in compact multiplicity form: e.g. 10×4 means four members of weight 10, a bare value means multiplicity one. The exact Banzhaf swing-count vectors β and the derived power distributions (percentages of total swings), in the same notation, are: EEC-6 $\beta = (10 \times 3, 6 \times 2, 0)$, power = $(23.8 \times 3, 14.3 \times 2, 0)\%$; EEC-9 $\beta = (53 \times 4, 29 \times 2, 21 \times 2, 5)$, power = $(16.7 \times 4, 9.1 \times 2, 6.6 \times 2, 1.6)\%$; EEC-10 $\beta = (100 \times 4, 52 \times 3, 26 \times 3)$, power = $(15.8 \times 4, 8.2 \times 3, 4.1 \times 3)\%$; EEC-12 $\beta = (286 \times 4, 242, 148 \times 4, 102 \times 2, 40)$, power = $(12.9 \times 4, 10.9, 6.7 \times 4, 4.6 \times 2, 1.8)\%$; EU-15 $\beta = (1849 \times 4, 1531, 973 \times 4, 793 \times 2, 595 \times 3, 375)$, power = $(11.2 \times 4, 9.2, 5.9 \times 4, 4.8 \times 2, 3.6 \times 3, 2.3)\%$.

† *Beyond the certified range ($n \geq 11$): exact infeasibility is not certified for these bodies (Tier-3 runs are heuristic, Section 9); the power distributions and L_1 values are exact computations reported as scaling context only, carrying no certificate.*

Here L_1 is the distance to ψ^n minimized over which member occupies the half-power position of the egalitarian-with-one-exception target. Three observations:

- **EEC-6 has a null player.** Luxembourg (weight 1, quota 12 of total 17) has zero swings: no coalition of the remaining five members sums to weight 11, so Luxembourg’s vote is never decisive. Its power is 0, while the three large members (Germany, France, Italy) jointly hold 71.4% of all power. This is a known pathology of the original weighted Council, reproduced here exactly.

- **Real power is far from egalitarian.** The three bodies sit at L_1 -distance 0.338–0.420 from ψ^n , and the power spread (largest/smallest positive power) ranges from $3.85\times$ (EEC-10) to $10.6\times$ (EEC-9).
- **All three lie in the certified range** ($n = 6, 9, 10$), so the obstruction of Section 9 applies to them directly (exact infeasibility; the distance values above are exact subset-enumeration computations, reproduced in the companion code package).

12.2 What the certified machinery tells a designer

For each body with $n \leq 10$, Section 9 certifies, deterministically, at the conclusive weight bound $w_i \leq 2^{n-1}$, that **no weighted majority game on n voters realizes ψ^n exactly**. The decision-support consequences for an institutional designer are:

1. **Certified infeasibility of the target.** No re-weighting of these councils, indeed no weighted rule on ≤ 10 voters at all, can attain the egalitarian target ψ^n . A designer who insists on a weighted rule must accept a strictly positive distance to ψ^n .
2. **Certified best achievable ($n \leq 8$).** Section 10.3 computes the exact minimal L_1 -distance over all weighted games: $5/44 \approx 0.114$ for $n = 6$ (attained at weights $(3,1,3,3,3,1)$, quota 11, and verified against exhaustive enumeration of all 1,111 weighted games), 0.0855 for $n = 7$, and 0.0667 for $n = 8$, each certified as a global optimum over the full total-swing range. The actual EEC-6 rule sits at $L_1 = 26/77 \approx 0.338$, so even within the weighted class, a redesign could improve the distance to ψ^6 by a factor of about three.
3. **Boundary of certification.** For $n \geq 11$ the exact infeasibility is not certified (Tier 3 runs are heuristic; Section 9). The later EEC-12 ($n = 12$) and EU-15 ($n = 15$) bodies lie beyond the certified range; their exact power distributions in Section 12.1 are reported as scaling context only, and their distance values carry no certificate.

12.3 The design decision and institutional context

The obstruction of Section 9 is a structural statement about the weighted class: egalitarian power is unattainable for $n \leq 10$, and the real EU bodies are far from it. This is consistent with the actual trajectory of the institution (**contextual**):

the Council moved away from a single weighted quota in the Lisbon Treaty, adopting a double-majority rule (at least 55% of member states representing at least 65% of the population) that cannot be represented as one weighted majority game. The certified obstruction supplies the formal decision-support complement to that historical choice. It quantifies how far weighted rules can be from an egalitarian target and certifies that they cannot reach it for small bodies; via the procedure of Section 10.3, it returns the exact best weighted approximation when the designer insists on staying in the weighted class.

12.4 A prescriptive design-analysis procedure

For a designer facing a target ψ on an n -voter body, the two engines of Section 10 compose into a decision-support procedure:

- if $n \leq 10$ and $\psi = \psi^n$, the Section 9 obstruction certifies the decision (infeasible in the weighted class; feasible by explicit construction in the general class, Section 5);
- if $n \leq 8$ and $\psi = \psi^n$, the Section 10.3 optimizer returns the certified best weighted approximation and its distance;
- otherwise the same solvers run under a time budget, and the outcome is labelled heuristic (Tier 3), not a certificate.

All numbers in this section are exact computations reproduced in the companion code package.

13. Conclusion

We complement the construction half with a negative result in the *weighted* universe (Section 9): no weighted majority game on six to ten voters realizes the egalitarian target ψ^n . The obstruction is anchored in a classical swap/block identity (Proposition 4) for swing-count differences in weighted majority games, which forces the distinguished player to carry minimal weight and concentrates every other player's deviation into a rigid subset-sum window. The sharp mirror form (M') of those windows is new, and it is our central structural contribution. Deterministic CP-SAT, at the conclusive weight bound $w_i \leq 2^{n-1}$, certifies infeasibility for $n = 6, 7, 8, 9$ and 10 ($n = 6$ consistent with the exhaustive enumeration of all 1,111 weighted games, whose optimal L_1 -distance to ψ^6 is $5/44$). The same computation certifies the quantitative form of Conjecture 15, the universal bound $\|\text{Bz}(v) - \psi^n\|_1 \geq c/n$, for $n = 6, 7$ and 8 , where the

optimal L_1 distances $5/44$, 0.085470 and 0.066667 give $c \leq 0.533$. The two-tier generalization $\psi^n(3, 1)$ is shown to be realizable by a *uniform* construction for all $n \geq 6$, while the triangle constructions of the W-family are obstructed for $\psi^n(k, 1)$ with $k \geq 4$. The two Kurz conjectures are therefore settled as follows: Conjecture 16 holds by an explicit construction valid for every $n \geq 6$, and Conjecture 15 holds in its quantitative form for $n \leq 8$ and in the exact case for $n \leq 10$, with the structural mechanism (M') explaining why. The general case is Open problem 4 of Section 11.

As a contribution to the computational study of voting rules, the paper offers a reusable, two-sided methodology rather than a one-off verification. Its construction engine (Section 10.1) is closed-form for every $n \geq 6$ and computationally verified for every $6 \leq n \leq 80$, and its exact verification engine (Section 10.2) certifies non-existence at a conclusive weight bound, with the structural property (M') as its pruning engine. Both halves are fully reproducible from the companion code package. The Section 12 case study grounds this methodology in a concrete institution, the weighted Council of the European Union, turning the abstract inverse-optimization problem into prescriptive decision support for the design of voting rules.

Appendix. Two numerical observations, and the case for exact formulations

The appendix records material that supports the main line without being needed for any proof.

Observation 2 (numerical). In all known solutions the scaling factor c is even ($c \in \{4, 6, 8, 10, 12, 14, 16, 22, 24, \dots\}$). This is not a feature of the recorded solutions alone: the swing counts of any simple game share a common parity, so for the target $(2c, \dots, 2c, c)$ the scale is forced to be even (Corollary 5).

Observation 3 (structure of the MILP witnesses). The MILP solutions for $n = 8, 10, 11, 12, 15$ (Section 7) all belong to the W-family, and their threshold parameters (G, F) share two features: the special player has low degree in G (1-3) and appears in most F -triples, and G is connected. These reflect the solver’s specific choices, and the recorded scale is itself solver-dependent (at $n = 43$, $c = 56$ under HiGHS and $c = 362$ under CP-SAT, since any multiple of the minimal feasible c is feasible). The closed forms cover these instances at every

n : Proposition 8 for $n \equiv 0, 2, 3, 6 \pmod{7}$, Proposition 13 and Corollary 15 for $n \equiv 1, 4, 5 \pmod{7}$, and Corollary 16 for $n = 10$.

Search is not a viable route to these constructions. For $n = 8$, which the closed forms settle, 4×10^6 monotone random (E, F) samples of the threshold family produced no realization: the realizing set has measure essentially zero in the sampled space. Random and heuristic search therefore do not reach these games at the sizes of interest; the exact formulations of Section 10 are what make them available, until the closed forms of Propositions 8 and 13 remove the solve entirely for every $n \geq 6$.

Declarations

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability. All results reported in this paper are reproducible from the companion code package (CP-SAT/HiGHS models, instance tables, and recorded runtimes). The version of that package which reproduces this manuscript is available from the corresponding author on request, and will be deposited in the project repository at

<https://github.com/chenyongjingg/banzhaf-inverse-power>

on publication.

CRedit authorship contribution statement. Yongjin Chen: Conceptualization, Methodology, Investigation, Writing — original draft. Ziyang Xue: Formal analysis, Software, Data curation, Writing — review & editing. Songze Zhu: Conceptualization, Supervision, Writing — review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the authors used Claude (Anthropic) to draft and revise text, including a new benchmark section, and to implement and run parts of the computational experiments reported in it, alongside language editing, typesetting, and formatting of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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